CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of vertices of a cuboid is: 1

(a) 8 **(b)** 6 **(c)** 12 **(d)** 4

2. In the Sierpinski triangle construction, the number of small shaded triangles at stage 4 (stage 0 has one) is: 1

(a) 80 **(b)** 48 **(c)** 81 **(d)** 12

3. The number of faces of a heptagonal prism is: 1

(a) 8 **(b)** 7 **(c)** 9 **(d)** 14

4. The number of vertices of a heptagonal pyramid is: 1

(a) 7 **(b)** 8 **(c)** 14 **(d)** 9

5. The number of edges of a octagonal prism is: 1

(a) 24 **(b)** 18 **(c)** 10 **(d)** 16

6. A polyhedron has 8 faces and 6 vertices. By Euler's formula, its number of edges is: 1

(a) 13 **(b)** 12 **(c)** 14 **(d)** 24

7. The sum of the lengths of all the edges of a cube of side 7 cm is (in cm): 1

(a) 42 **(b)** 28 **(c)** 84 **(d)** 56

8. A solid with a polygon as base and triangular faces meeting at a point is a: 1

(a) cuboid **(b)** prism **(c)** cylinder **(d)** pyramid

9. In the Sierpinski triangle construction, the number of small shaded triangles at stage 5 (stage 0 has one) is: 1

(a) 15 **(b)** 243 **(c)** 242 **(d)** 96

10. The number of faces of a octagonal prism is: 1

(a) 8 **(b)** 10 **(c)** 16 **(d)** 9

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the total length of all edges of a cube of side 11 cm. 2

12. Write the number of faces, vertices and edges of a octagonal prism. 2

13. A solid has 8 faces, 12 vertices and 18 edges. Verify Euler's formula. 2

14. Find the total length of all edges of a cube of side 15 cm. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. State the number of faces, vertices and edges of a heptagonal pyramid and verify Euler's formula. 3

16. In the Sierpinski triangle, how many shaded triangles are there at stage 6? How many more are there at stage 7? 3

17. A cuboid measures 9 cm by 7 cm by 2 cm. Find the total length of its edges and state its number of faces and vertices. 3

OR

State the number of faces, vertices and edges of a octagonal pyramid and verify Euler's formula. 3

18. In the Sierpinski triangle, how many shaded triangles are there at stage 7? How many more are there at stage 8? 3

OR

A cuboid measures 6 cm by 6 cm by 5 cm. Find the total length of its edges and state its number of faces and vertices. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A Koch curve begins with a straight line of length 45 cm (stage 0). At each stage every segment is replaced by 4 segments each one-third as long. Find the number of segments, the length of each segment and the total length at stage 2. 5

OR

In the Sierpinski triangle, stage 0 has 1 shaded triangle. Fill in the number of shaded triangles at stages 5, 6 and 7. By what factor does the count grow at each stage? 5

20. A Koch curve begins with a straight line of length 54 cm (stage 0). At each stage every segment is replaced by 4 segments each one-third as long. Find the number of segments, the length of each segment and the total length at stage 2. 5

OR

In the Sierpinski triangle, stage 0 has 1 shaded triangle. Fill in the number of shaded triangles at stages 6, 7 and 8. By what factor does the count grow at each stage? 5

END OF QUESTION PAPER Total: 40 Marks